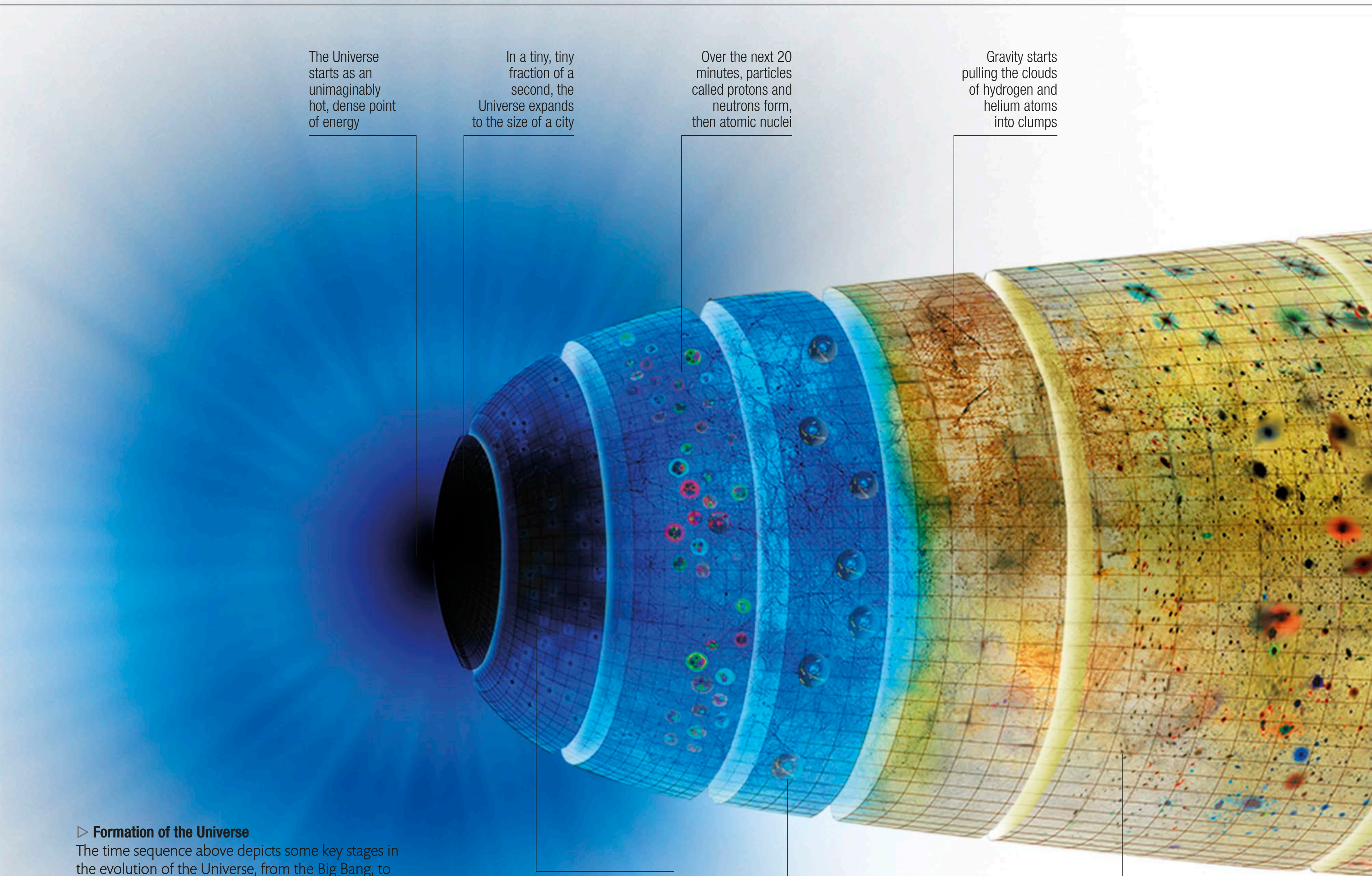
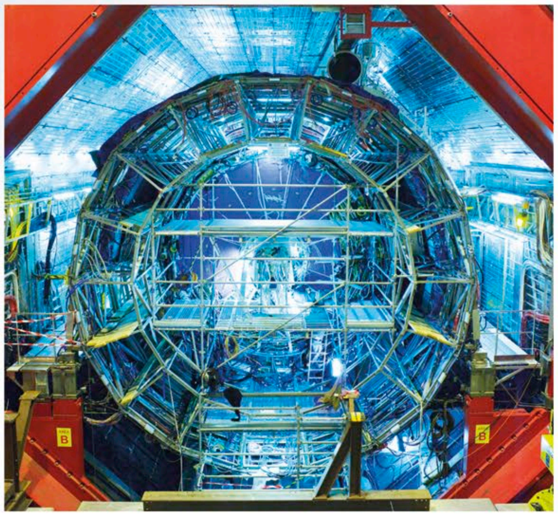


▷ **Formation of the Universe**
The time sequence above depicts some key stages in the evolution of the Universe, from the Big Bang, to the formation of atoms, then stars and galaxies, and events through to the present day and into the future. Since the Big Bang, the Universe has cooled and grown larger through the expansion of space itself.

THE BIG BANG

ABOUT 13.8 BILLION YEARS AGO, AN EXCEEDINGLY DRAMATIC EVENT MARKED THE BEGINNING OF BOTH SPACE AND TIME. FROM NOTHING, THE UNIVERSE SUDDENLY APPEARED AS A TINY POINT OF PURE ENERGY.

Within an instant, in what is known as the Big Bang, the Universe expanded trillions of trillions of times, and then continued to get larger, at the same time cooling from its stupendously hot birth. During the first fractions of a second, a vast "soup" of tiny, interacting particles formed out of the intense energy. Some of these joined to make the nuclei (centers) of atoms—the building blocks of everything we can see in the Universe today. Tens of thousands of years later, actual atoms formed and then, after hundreds of millions of years, the very first stars and galaxies.



◁ **Studying the Big Bang**
Using this complex machine, the Large Hadron Collider, scientists at the European Center for Nuclear Research (CERN) attempt to re-create the conditions that followed the Big Bang. In the collider, beams of high-energy particles are smashed together and the by-products studied.